



AI – Powered Community Disaster Resilience Platform for Sri Lanka

MINI HACKATHON 2026

Round 1 – Project Proposal

Team NovaForge

Topic Category – Society, Work & Accessibility

Selected Topic – Disaster Preparedness & Resilience

Sri Lanka Institute of Technology (SLIIT)

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1. Topic Declaration & Team Overview

Selected Topic – Disaster Preparedness & Resilience (Society, Work & Accessibility)

WarnX is an AI powered community disaster resilience platform designed to improve how communities prepare for, respond to, and recover from disasters.

The platform focuses on four stages of disaster management:

Predict → Alert → Community → Recover

WarnX is designed to complement existing official warning systems by providing localized risk intelligence, personalized alerts, AI assisted community verification, and coordination tools that connect residents, volunteers, and response organizations.

The WarnX Concept

Instead of only asking:

"Is there a disaster warning?"

WarnX aims to answer:

"Am I at risk? What should I do? What is happening around me? And who can help?"

Team Members

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2. Problem Statement

300 + Lives lost annually to floods and Landslides	2 hrs. Average disaster warning time in Sri Lanka	Most Of warnings are received after impact has already begun	To the best of our knowledge, no widely adopted AI powered community disaster platform currently exists in Sri Lanka
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Source: Disaster Management Centre (2024)

Sri Lanka already has established disaster warning mechanisms, including the Disaster Management Centre (DMC) and Disaster and Emergency Warning Network (DEWN). However, there are opportunities to improve the connection between official warnings and real time community level conditions.

- **Alerts are only one-way** DEWN & DMC send SMS broadcasts with no area limitation. No AI - same climate message to a fisherman in Matara and a farmer in Anuradhapura.
- **No local predictions** Existing systems provide district level warnings. A village 2 km from a flooding river gets the same warning as 20Km away people get the same warning.
- **No communication** Rising water levels spotted by residents are not aware from official systems. Communities cannot coordinate help or share real time situations.
- **Lack of connectivity and literacy** No existing tool works offline or in all three officials’ languages with voice support.

Despite existing tools like DEWN, preventable losses continue due to the lack of community intelligence and communication problems.

3. Proposed Solutions – The WarnX application Blueprint

WarnX is structured around the full disaster journey – prediction, alerting, community coordination and recovery. So, this single platform guides people from the first sign of risk to post disaster support.

1. PREDICT - Early warning & Risk Intelligence

- Local AI Risk alerting: AI model predicts floods, landslides, and drought risks at Grama Niladri (GS) Division level.
- Real time integration with DMC official APIs, OpenWeatherMap, and satellite imagery (e.g. - rainfall, river water levels).
- The AI prediction model will be trained on historical rainfall, flood records, weather forecasts, and geographical data to generate localized, advance-notice disaster risk predictions.
- **User Benefit:** Villagers receive personalized risk alerts before water starts rising, giving them critical time to prepare.

2. ALERT - Smart Notifications

- Push notifications and SMS for no internet areas, voice alerts in Sinhala, Tamil & English
- Alerts are personalized with clear instructions based on users' location and family background (e.g. - Evacuate now via safe route. Nearest shelter is 1.2 km away").
- Color coded alerts (Low / Medium / High / Extreme) with simple icons and maps.
- pre-loaded risk maps and last alerts work without internet.
- **User Benefit:** Even low connectivity & literacy users in rural areas understand and act immediately.

3. COMMUNITY CONNECTION - Coordination

- **Crowdsourced Reporting:** photo, video, description upload for incidents (rising water, blocked roads, damaged bridges).
- **AI Verification** - Filter out the fake news and highlight the verified news □ Interactive map that displays the live reports blocked roads Damage bridges based on confirm news.
- **Help Matching Board:** "I need help" (e.g., boat, medicine) matched with "I can offer" posts using the Coordination Agent.
- **User Benefit:** Turns passive residents into active contributors - local situations reach officials faster and neighbors help each other.

4. RECOVER (Post disaster support)

- Easy sends the current location when need help in one click to responsible authorities.
- **Damage Reporting:** Easy form with photos for submitting home/crop damage to government and NGOs for faster aid.
- **Recovery Dashboard:** Track progress of road clearing, power restoration, and medical camps.
- Recover guidance through the AI agent with easy routes to hospitals and medical camps based on verified information
- **User Benefit:** Helps families recover faster and builds them to future disasters.

4. AI Agent Architecture

WarnX is powered by a coordinated multi agent system built with LangGraph and LangChain, where each agent owns a distinct responsibility across the disaster lifecycle.

- **Weather Monitoring Agent** - continuously watches available weather and disaster-risk data sources, checking at short, regular intervals to detect early signs of danger and issue localized risk warnings ahead of an event.
- **Action Suggestion Agent** - functions as a personal advisor, referencing Sri Lankan disaster-response guidelines, known shelter locations, and historical flooding areas to generate practical, situations specific guidance for each house.
- **Coordination Agent** - acts as a community coordinator: it collects incoming reports (photos, videos, descriptions), assesses their reliability, and matches people who need help with people or authorities who can provide it enabling neighbors to help each other quickly rather than waiting solely on official response.
- **Learning Agent** - reviews outcomes after each event real disaster areas, community reports, and response effectiveness and feeds those learnings back into the system so predictions and guidance improve with every cycle.

Together, these four agents let WarnX get measurably smarter and safer over time, rather than functioning as a static alerting tool.

Responsible AI & Safety

Because WarnX operates in a disaster response context, AI-generated information must be treated carefully.

WarnX will not replace official warnings or emergency authorities. AI-generated predictions and recommendations will act as supporting information.

Community reports will be treated as unverified until sufficient supporting evidence is available.

For AI-generated guidance, WarnX will prioritize trusted disaster-response information and a controlled knowledge base to reduce unsupported or misleading recommendations.

5. Technical Architecture & Feasibility

Layer	Technology	Why We Chose It
Mobile App	React Native + NativeWind	Single codebase for Android and iOS; fast, consistent development across the team.
Backend API	Python + FastAPI	High performance, strong AI/ML ecosystem support, well suited to real-time alert handling.
AI / Agents	LangGraph + LangChain, OpenAI API	Purpose-built for orchestrating multiple cooperating AI agents.
Database	PostgreSQL + Redis	PostgreSQL's PostGIS support handles geographic data reliably; Redis supports fast real-time operations.
Maps / Offline	MapLibre + OpenStreetMap	A fully free, open mapping solution with offline tile support — well suited for rural, low-connectivity areas.
Push & SMS	Firebase Cloud Messaging + Twilio (SMS fallback)	Reliable delivery even in areas with poor network coverage.
Hosting	Railway	Straightforward, scalable deployment suited to a growing student-led project.
Data Integrations	OpenWeatherMap (confirmed); DMC public data feed (pending confirmation of public API availability)	OpenWeatherMap provides an immediately usable meteorological feed; direct DMC integration will be pursued through official channels once access is confirmed.

Minimum Viable Product

To ensure feasibility within an undergraduate development environment, the initial WarnX prototype will focus on the following core capabilities:

- User location and interactive risk map
- Weather data integration
- Localized risk-level calculation
- Push notifications
- Community incident reporting
- AI-assisted report analysis
- Basic community helps match

Advanced features such as large-scale government integration, advanced prediction models, extensive offline mapping progressively developed after the core system is validate

6. Target Audience & Market Opportunities

22M+	10M+	2600+	To the best of our knowledge, no AI powered community disaster platform currently exists in Sri Lanka
Sri Lankans within potential reach of WarnX	People living in high-risk rural areas	Grama Niladri divisions addressable with localized alerts	

Primary users of the WarnX are residents of flood and landslides communities across Sri Lanka. Particularly in rural areas Grama Niladri divisions were warnings precision and offline reliability matter most. **Secondary users** are our community as Sri Lankans we are always together in a disaster situation. So, all the other people and volunteers, local authorities and NGO field teams are also the users of WarnX

7. Market Strategy & Sustainability

Market Strategy

- **Government partnership** - engaging the Disaster Management Centre to explore WarnX as a complementary community feature alongside the existing DEWN system, which could extend reach to millions of users quickly.
- **NGO collaboration** - partnering with organizations such as Sarvodaya and World Vision Sri Lanka to support installation and community training in villages.
- **Organic community growth** - disaster apps naturally spread through word of mouth during real events; this will be paired with proactive awareness programs in flood areas.

Sustainability

- **Revenue model** - free for individual/community users; premium analytics dashboards offered to NGOs and government offices.
- **Funding** - pursuing grants from organizations focused on disaster resilience, such as UNDP and the Asian Development Bank.
- **Scalability** - the same architecture can be adapted for other disaster occurring countries such as Bangladesh, Nepal, or Myanmar with regional configuration changes.
- **Compounding value** - the Learning Agent means the platform's predictions and guidance improve as adoption grows, reinforcing long term relevance.

Revenue model expansion

- **Government subscription** - Regional and national disaster management offices subscribe for access to aggregated, real-time risk dashboards across GS Divisions, supporting faster resource allocation decisions during active events.
- **NGO analytics tier** - Organizations like Sarvodaya or World Vision pay for historical and live incident data to plan relief logistics and target community training more precisely.
- **Disaster-risk API access** - Insurance companies and reinsurers license access to WarnX's localized risk-prediction data to inform flood and landslide risk assessments for policyholders.
- **Enterprise reporting dashboard** - Large NGOs and corporate CSR programs pay for a customized reporting view to track their own disaster-response impact and outcomes.

8. Competitive Advantage

Feature	WarnX	DEWN	Other Existing Apps
Local AI prediction	GS Division level	District only	None
Community crowdsourcing	AI-verified reports	None	Basic
AI agent system	4 coordinated agents	None	None
Offline support	Full offline mode	SMS only	None
Community help matching	AI-powered board	None	None

DEWN is primarily designed for broad emergency dissemination. WarnX approaches the problem from a community-intelligence perspective, combining localized risk assessment, two-way community reporting, and AI-assisted coordination. Rather than replacing DEWN, WarnX is designed to complement existing official warning infrastructure with a more localized and interactive layer

WarnX will one of best platform in Sri Lanka that combines localized AI prediction, verified community crowdsourcing, and coordinated multi agent support across the full disaster lifecycle before, during, and after an event.

9. Development Roadmap

- **Phase 1 - Core Alert App:** React Native app shell, real-time push alerts, and an interactive risk map, alongside initial data integrations.
- **Phase 2 - AI Agent Layer:** Weather Monitoring and Action Suggestion agents, a RAG-based knowledge base, and crowdsourced incident reporting.
- **Phase 3 - Coordination & Recovery:** Coordination and Learning agents, the recovery dashboard, one-tap location sharing, and multilingual voice alerts.
- **Phase 4 - Hardening & Launch:** Device testing, error handling, UX refinement, and production hosting.

10. Conclusion

WarnX addresses an important gap in community disaster resilience by connecting official information, environmental data, and real-time community intelligence.

Rather than functioning only as a warning application, WarnX supports the complete disaster journey:

PREDICT → ALERT → COMMUNITY → RECOVER

The platform is designed to complement existing disaster-management systems by providing localized risk intelligence, personalized guidance, AI-assisted community verification, and coordination tools.

As an undergraduate innovation project, WarnX is designed around a realistic MVP that can be demonstrated and progressively expanded toward real-world deployment.

WarnX

Turning disaster information into community action.